

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 12: Surface Areas and Volumes

Previous year board questions, topic-wise: 11 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Surface area

- Find the total surface area of a solid hemisphere whose diameter is $2d$. (a) $3\pi d^2$ (b) $2\pi d^2$ (c) $\frac{1}{2}\pi d^2$ (d) $\frac{3}{4}\pi d^2$ **CBSE 2026, 1 mark**
- Assertion (A): Two identical solid hemispheres joined along their flat faces form a sphere. Reason (R): The total surface area of a sphere of radius r is $3\pi r^2$. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2025, 1 mark**
- A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2 **CBSE 2024, 1 mark**
- A cylinder 5 cm high has a curved surface area of 94.2 cm^2 . Find its radius, taking $\pi = 3.14$. (a) 2 cm (b) 3 cm (c) 2.9 cm (d) 6 cm **CBSE 2023, 1 mark**
- Find the curved surface area of a cone of height 24 cm and radius 7 cm. (a) 528 cm^2 (b) 1056 cm^2 (c) 550 cm^2 (d) 500 cm^2 **CBSE 2023, 1 mark**

2 Volume

- A solid is a cylinder with a hemisphere on each end. The solid is 20 cm long overall and the cylinder is 7 cm in diameter. Find the volume of the solid. Take $\pi = \frac{22}{7}$. **CBSE 2026, 3 marks**
- A room has the shape of a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21} \text{ m}^3$ of air. Find the height of the cylindrical part. Take $\pi = \frac{22}{7}$. **CBSE 2025, 3 marks**
- The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a) $\frac{4\pi}{3} \text{ cu cm}$ (b) $\frac{5\pi}{3} \text{ cu cm}$ (c) $\frac{8\pi}{3} \text{ cu cm}$ (d) $\frac{2\pi}{3} \text{ cu cm}$ **CBSE 2024, 1 mark**
- A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm^3 . Find the two radii. **CBSE 2024, 3 marks**
- A room is a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21} \text{ m}^3$ of air. Find the total height of the room. Take $\pi = \frac{22}{7}$. **CBSE 2023, 3 marks**

11. An ice-cream cone has radius 3 cm and height 12 cm. The lowest part of the cone, one-sixth of its volume, is left empty. The rest is filled with ice-cream, with a hemisphere of ice-cream on top. Find the volume of the ice-cream. Take $\pi = 3.14$. **CBSE 2023, 3 marks**

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